Project AI/ML Travel Price Advisor

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Annotation

Tento project is designed as an intelligent assistant for travellers looking for the best and cheapest time to purchase flight, bus, or train tickets. Based on a machine learning model, the application predicts whether the current price of a connection is advantageous, average, or above average, and recommends a specific course of action to the user. Data for training the model was collected over several weeks using custom scrapers from publicly available sources: Ryanair API (flights), RegioJet API (buses and trains), Open-Meteo API (weather) and Nager.Date API (holidays and school breaks). For now a total of over 15,000 records were gathered, covering 20 air routes and 8 ground routes from Prague to various European destinations. The model was trained as an XGBoost classifier with three output classes (BUY_NOW, WAIT, EXPENSIVE) and achieves an accuracy of approximately 75%. Resulting web page is build on the Streamlit framework.

Key words: AI/ML, Python, classification

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1. Introduction

The goal of the project is to create an application that uses personally collected data to predict whether a given price is a bargain or overpriced, advising the user on when and how to travel most affordably.

The project consists of four main parts:

- **Data collection:** four scrapers collecting data from Ryanair, RegioJet, Open-Meteo, and public holidays
- **Preprocessing:** cleaning, encoding, and scaling data in Google Colab
- **Model:** training the XGBoost classifier and exporting it into .pkl files
- **Application:** a Streamlit web application for the end userTechnický rozbor

SymptomTracker je aplikace, která funguje offline - nepotřebuje přístup na internet pro základní používání. Veškerá data (záznamy a fotky) se ukládají přímo v paměti telefonu.

2. Technology

Project written in Python.

- Data collection: request, cloudscraper
- Data processing: pandas, scikit-learn
- Model: XGBoost
- Web interface: Streamlit

2.1.Ryanair scraper

The scraper uses the cloudscraper package to bypass Cloudflare protection. It scans 20 routes from Prague (PRG) to European destinations, checking prices for 60 different dates over a 120-day horizon

2.2. RegioJet scraper

This scraper downloads prices for 8 routes. A specific challenge was the duration format (e.g., '18:30 h'), which was processed by a custom parse_duration function to convert text into minutes.

3. Data preprocessing

3.1. Cleaning

Data types were corrected (text to datetime or numeric). Records with missing key values were removed, along with 1 duplicate and 123 price outliers using the IQR method.

3.2.New features

- day_of_week
- month
- departure_hour
- is_holiday

3.3. Target variable - Label

The model predicts a price class based on the deviation from the average for that route and transport type:

Label	Condition	Data Proportion
BUY_NOW	deviation < -20%	~48%
WAIT	-20% to +20%	~25%
EXPENSIVE	deviation > +20%	~27%

4. Machine learning model

4.1. Configuration

XGBoost was configured with 300 trees, a maximum depth of 6, a learning rate of 0.05, and a 0.8 subsample. Data was split 80/20 for training and testing.

```
Best parameters:
{'subsample': 0.8, 'n_estimators': 400, 'max_depth': 6,
'learning_rate': 0.05}
```

5. Application development

5.1. Architecture

- **Collection Layer:** Python scrapers
- **Analytical Layer:** Google Colab for training
- **Model Layer:** Exported `.pkl` files and advice.json
- **Presentation Layer:** Streamlit

5.2. Project structure

- scraper/: scraper scripts for flights, RegioJet, weather, and holidays
- model/: trained model, scaler, label encoder, and advice.json

- app.py: the Streamlit web application

6. Execution

1. Install Requirements: pip install -r requirements.txt

2. Dowlnoad project from GitHub

3. Open in PyCharm

4. Run Scrapers: run individual scripts in the scraper/ folder

5. Train Model: run all cells in the Google Colab notebook.

6. Launch App: streamlit run app.py

7. Conclusion

The project successfully met all criteria, creating a functional system for multi-source data collection and price prediction. With an accuracy of 75%, it significantly outperforms a random guess. Future expansions could include accommodations as parameter and popularity of travels for better predictions.

8. Contact

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